

LED FILAMENT BULBS Growing Your Business with Aesthetics



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A jeweller recently replaced the traditional incandescent lights of his store with LED filament bulbs. Ask him why he did it and pat comes the reply: "I wanted my diamonds to shine bright and look attractive to the customers."

LED filament bulbs are taking the market by storm. Enterprises in hospitality, lifestyle and jewellery sectors have been increasingly replacing their Edison bulbs (the regular incandescent bulbs that were invented by Thomas Edison) with this new kind to enhance their ambience and get more customers.

What makes LED filament technology special

The LED filament technology is quite simple, using a chip-on-glass and a particular mixture of gases inside the glass bulb.

The chip-on-glass technology uses a

glass substrate instead of the traditional metal substrate to mount unpackaged LEDs, which helps the LED filament bulbs to emit light uniformly without losses. Once LED chip bonding is done, it is encapsulated in a phosphor material, which affects the colour of the light emitted by the bulb. Some manufacturers also use this phosphor for temperature control of the product.

Vinod Kumar, vice-president, R&D, at Surya Roshni, shares that the circuit and driver technology has not changed so far. "Design engineers had to work really hard with the technology to make it an electromagnetic-compatible product in the same form factor, which results in lower fluctuations," he adds.

The proportion of gases inside the glass bulb decides the temperature of the product. This, in turn, directly affects the longevity of the bulbs. LED majors, however, rely more on the high-voltage,

